

MGMT 205

Session 4

Saving for Retirement, Social Security, and Time Value of Money

Presented by

Mary Sasmaz, PhD, CPA, CFP



Before we start....

- 💧 ID Theft Protection – S.T.O.P method

<https://www.youtube.com/watch?v=yxlR1st8QKI>

Fact or Myth? Social Security won't be around by the time you retire.

0

A Fact

0%

B Myth

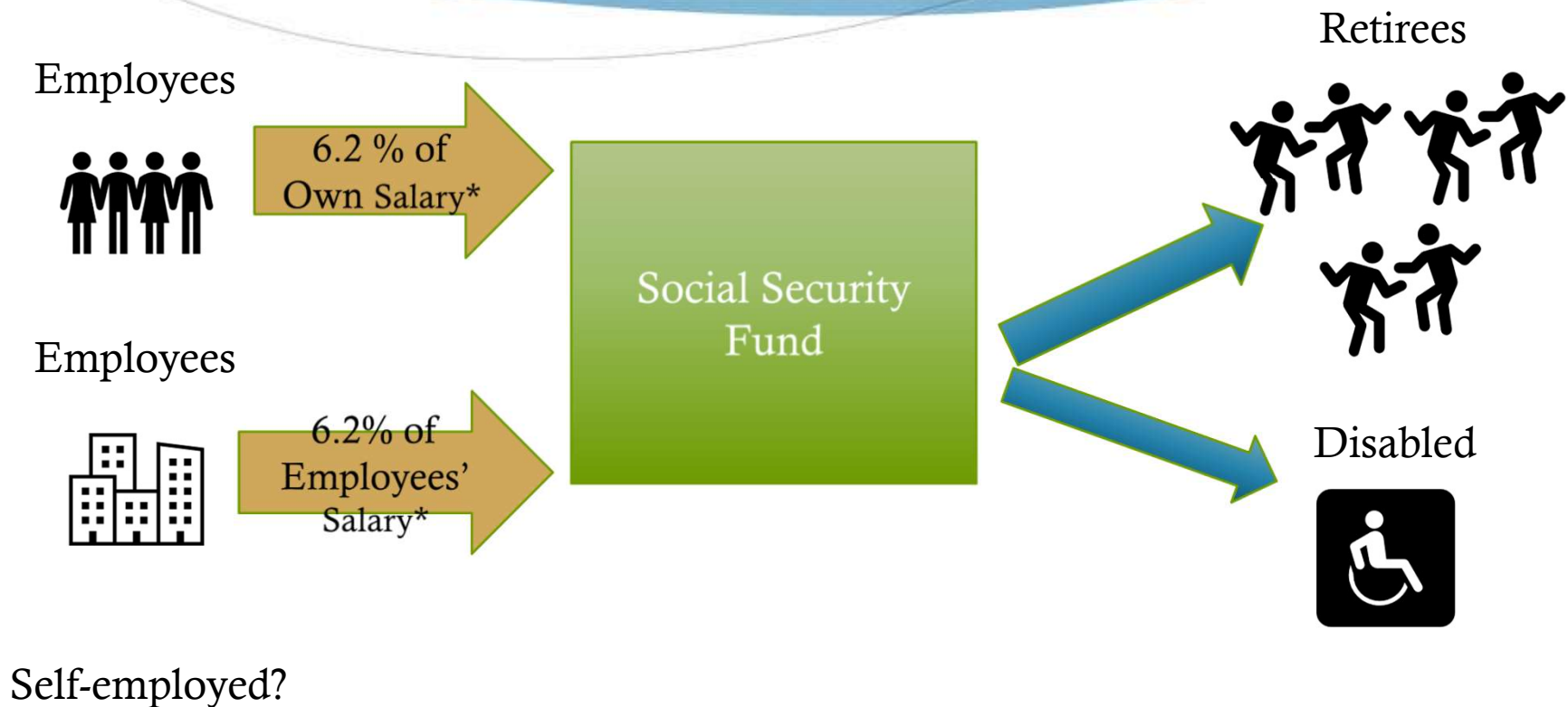
0%

Fact or Myth?

Social Security won't exist by the time you retire?

MYTH!!!

Social Security



* Contributions on salary up to \$176,100 in 2025. Indexed for inflation.

To be eligible for Social Security Retirement Benefits

- ◆ 40 Credits
 - ◆ 1 credit per \$1,810 of income in 2025
 - ◆ 4 credits max per year (\$7,240 income for year)
- ◆ Early Distributions – Age 62
- ◆ Full Retirement – Based on birth year (mine is age 67)
- ◆ Retirement Benefits are based on earnings during highest 35 years of reported earnings. (0 in a year with no earnings).

Social Security (SSA.gov)

Age to receive full Social Security benefits

<i>Year of birth</i>	<i>Full retirement age</i>
1943-1954	66
1955	66 and 2 months
1956	66 and 4 months
1957	66 and 6 months
1958	66 and 8 months
1959	66 and 10 months
1960 and later	67

NOTE: People born on January 1 of any year, refer to the previous year.

Retirement Income Sources

- ◆ Social Security
- ◆ Pension (not common anymore)
- ◆ Employer-sponsored savings programs (401k, profit sharing, etc.)
- ◆ IRAs
- ◆ Taxable Savings/Investment Accounts

Social Security

- ◆ Administered for you by the SSA
- ◆ Contributed to by both employer and employee
- ◆ Payments adjusted for inflation annually
- ◆ Transparent – Check your estimated benefit at [SSA.gov](https://www.ssa.gov)
- ◆ REALITY
 - ◆ Will only replace between 20% and 45% of your income, though more recent studies suggest to expect closer 20%
 - ◆ Cannot pass on to your heirs (exception: portion to spouse)
 - ◆ Current estimates suggest it can only pay 78% of “promised benefit” starting 2034.

Pensions

- ◆ Administered by employer
- ◆ Generally replaces between 25-50% of your income
- ◆ Employer contributes 100%
- ◆ REALITY
 - ◆ Less than 25% of employers offer a traditional pension
 - ◆ Favors long-term employees over short-term employees
 - ◆ A material number of pension plans are “underfunded” and may not be able to pay 100% of promised benefits to current and future employees.

Employer-Sponsored Savings Plans

- ◆ Contributions are tax deductible (i.e. “pay less in taxes”)
- ◆ Employer match (i.e. “free money”)
- ◆ Administered by the employer – Automated Savings!
- ◆ REALITY
 - ◆ Employee contributes and has to make investment decisions
 - ◆ Withdrawals from plan before age 59.5 subject to taxes and penalties
 - ◆ Investment options are limited, and plan expenses may be high

IRAs

- ◆ Choice of **Traditional IRA** or **Roth IRA**
- ◆ Easy to set-up
- ◆ Large number of investment options
- ◆ REALITY
 - ◆ 100% funded by employee
 - ◆ Employee must make investment decisions and manage portfolio
 - ◆ May not be tax deductible due to income limitations
 - ◆ Subject to Congressional and IRS regulation

Saving for Retirement



Retirement

How much should I save?

💧 IT DEPENDS!

RETIREMENT

How much should I save?

- ◆ What kind of **Lifestyle** do you want in retirement?
- ◆ How large must your retirement be to fund your desired retirement?
- ◆ How much will social security and pension payments contribute to your retirement fund?
 - ◆ You must save any difference or “short-fall” between need and SSA/Pension payments

Retirement

How much should I save?

- 💧 Lifestyle choice

- 💧 Same lifestyle that you have when you work?

- 💧 70% to 80% of pre-retirement gross income **

- 💧 Lower lifestyle than you have when you work?

- 💧 50% to 65% of pre-retirement gross income

- 💧 Higher lifestyle than you have when I work?

- 💧 85% to 100% of pre-retirement gross income

- 💧 ** This works because some expenses will disappear when you retire. Amounts are estimated though, actual results can differ.

Retirement Needs

Expenses that are expected to disappear / reduce when you retire		
	Range (% of income)	
FICA Taxes (6.2% + 1.45%)	7.65%	7.65%
Savings	5.00%	20.00%
Cost of Working	5.00%	10.00%
Mortgage Payments**	<u>10.00%</u>	<u>30.00%</u>
Total Reduction in Monthly Expenses	27.65%	67.65%

**** Important Assumption that mortgage is paid off by the time you retire.**

***Above amounts are estimates – actual amounts depend on individual circumstances.

How much should I Save?

Alternative Approaches

- ◆ Online Needs Assessment Calculators
- ◆ Do you want to only live on income and leave your principal savings to your heirs?

Retirement

How much to Save?

- ◆ Assumptions of the following savings plan:
 - ◆ Begin to collect Social Security Retirement Benefits at FULL retirement age
 - ◆ You do not qualify for pension benefits from an employer
 - ◆ Your retirement portfolio earns an annual nominal rate of return of 7.5% or higher.

Target Annual Savings Rate for Retirement

AGE 65 – 80% of average pre-tax income

Targeted % of Income	
Age	Savings Ratio
25	12%
30	12%
35	12%
40	12%
45	15%
50	15%
55	15%
60	15%
65	15%

EXAMPLE – TARGET SAVINGS

AGE 65 – 80% of average pre-tax income

Targeted % of Income		Examples
Age	Savings Ratio	Gross Income \$40,000 / \$50,000
25	12%	\$4,800 / \$6,000
30	12%	\$4,800 / \$6,000
35	12%	\$4,800 / \$6,000
40	12%	\$4,800 / \$6,000
45	15%	\$6,000 / \$7,500
50	15%	\$6,000 / \$7,500
55	15%	\$6,000 / \$7,500
60	15%	\$6,000 / \$7,500
65	15%	\$6,000 / \$7,500

FREE MONEY!!!!

- ◆ Many employers provide a company match to employees' contributions to the company retirement plan (401k, 403b, etc.)
- ◆ Match examples:
 - ◆ 100% match on up to 4% of salary contributed by the employee
 - ◆ 50% match on up to 6% of salary contributed by the employee
 - ◆ 25% match on up to 6% of salary contributed by the employee

Example

- ◆ Jenny just started a new job. She is eligible to participate in the company's retirement plan as of Day 1 (some employers require a 1 year wait). Her starting salary is \$50,000. Her employer matches 50% of employee contributions on up to 6% of salary.
- ◆ How much should Jenny save if she wants to save 15% of her salary toward retirement?
- ◆ Enter your response into POLL EVERYWHERE.

Start the presentation to see live content. For screen share software, share the entire screen. Get help at pollev.com/app

Example

Jenny's Annual Salary\$	\$50,000
Saving Desire	15%
Max Company Match (50% of 6%)	3%
Saving Need	12%

How much is that?	
Annual Salary	\$50,000
Annual Retirement Savings	$50,000 \times .12 = \$6,000$
Monthly Salary	$50,000 / 12 = \$4,166.67$
Monthly Retirement Savings	$6,000 / 12 = \$500$

Example to Save 12 – 15% of Salary for Retirement

401k Plan Match	Comments
0%	You must save 12-15% annual to meet goal.
1%	With a company match of 1% of your salary, you must save an additional 11-14% to meet goal.
3%	With a company match, you must save an additional 9-12% to meet the goal.

Review of Assumptions

What if?

Assumption	If	Then
You retire at SSA full retirement age	Retire Earlier	Must Save MORE
	Retire Later	Can Save Less
Live on 80% of Gross Pre-tax Income	Want / Need More	Must Save MORE
	Live on Less	Can Save Less
Earn annual average nominal return on retirement investment of 7.5%	Earn Lower Rate	Must Save MORE
	Earn Higher Rate	Can save LESS

Retirement Savings Goal On Track?

AGE 65 – 80% of average pre-tax income

Age	Goal Multiple of Gross Pre-tax Income
25	.1
30	.6
35	1.4
40	2.4
45	3.7
50	5.2
55	7.1
60	9.4
65	12

Example – Retirement Savings Goal

AGE 65 – 80% of average pre-tax income

Age	Goal Multiple of Gross Pre-tax Income	Example Gross Income \$40,000 / \$50,000
25	.1	\$4,000 / \$5,000
30	.6	\$24,000 / \$30,000
35	1.4	\$56,000 / \$70,000
40	2.4	\$96,000 / \$120,000
45	3.7	\$148,000 / \$185,000
50	5.2	\$208,000 / \$260,000
55	7.1	\$284,000 / \$355,000
60	9.4	\$376,000 / \$470,000
65	12	\$480,000 / \$600,000

Retirement Savings Goals (Capital-to-Income Ratio)

	Retire at 65 with 80% of income	Retire at 65 with 70% of income	Retire at 70 with 70% of income
AGE	Multiple of Gross Pre-Tax Income		
25	.1	.1	.1
30	.6	.5	.5
35	1.4	1.3	1.0
40	2.4	2.0	1.5
45	3.7	3.1	2.5
50	5.2	4.5	3.5
55	7.1	6.1	4.8
60	9.4	8.1	6.5
65	12	10	8.2
70			9.25

So HOW do I do it?



Create a Budget

Budgeting Best Practices:

Spending: 50 / 20 / 30

Housing:

PITI at less than 20% of monthly gross income.

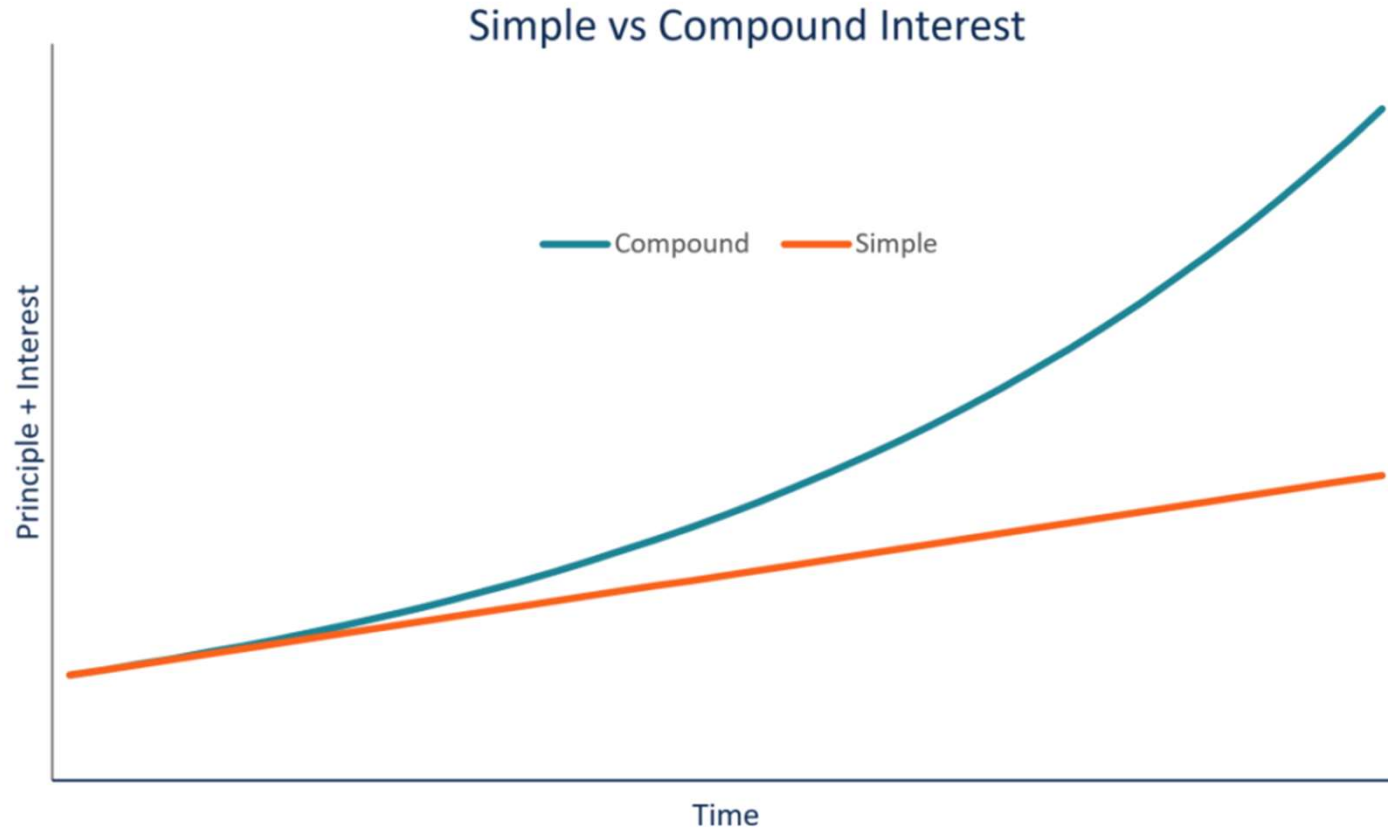
Housing + Utilities + Maintenance <30% of monthly gross income.

Transportation:

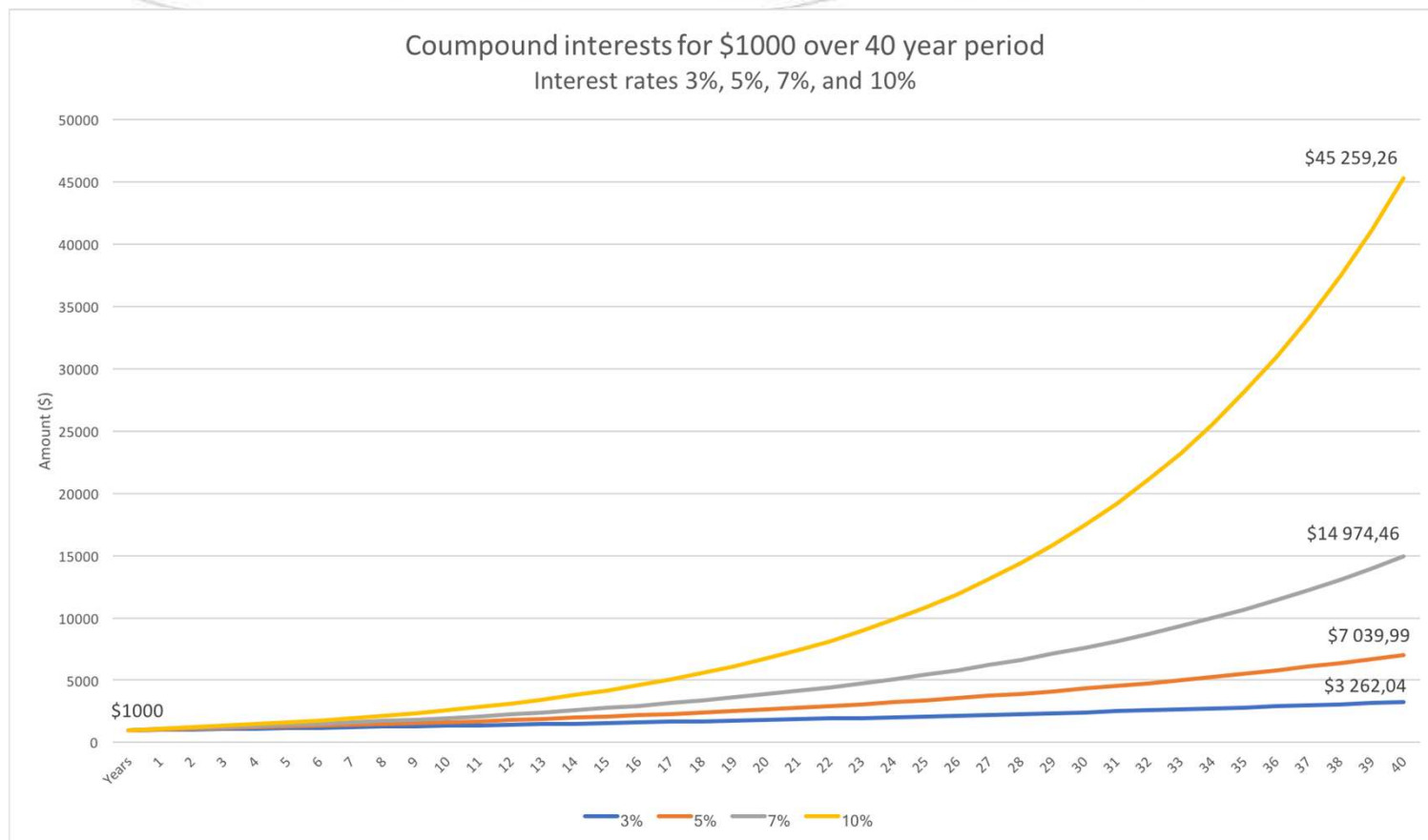
20/4/10 (20% down, 4-year max on financing, 10% gross income)

Compounded Interest

What is it?



Compounded Interest Example



Rule of 72

- ◆ Use to determine how long it takes to double your money.
- ◆ How long?
 - ◆ Divide 72 by your expected return to find out.
 - ◆ Example: Expected return of 6%, you will double your money every 12 years. ($72/6=12$)
 - ◆ Example: Expected return of 7.5%, you will double your money every 9.6 years. ($72/7.5=9.6$)

Example: Who saved more?

William	Frank
• Decides to start saving \$200/month toward Retirement • Invests money and earns annual average of 8.5% a year • 10 years from now stops contributing • Average rate of return rate continues for 30 years	• Today, purchases a home entertainment system, financed with monthly loan payment is \$200/month for 10 years. • 10 years from now, starts contributing \$200 a month to retirement account, earns average return of 8.5% • Invests \$200/month over next 30 years.

Example: Who saved more?

William	Frank
• Contributed total of \$24,000 (120 months x \$200) • Benefited from the “Miracle of Compound Interest” • His money was invested for longer period of time than Frank’s money. • End of 40 years: \$477,589.28	• Contributed total of \$72,000 (360 months x \$200) • End of 40 years: \$330,141.14

Another Example

Accumulated Amounts							
<i>Flossie—\$5,000 annual contributions for 10 years; \$50,000 total contributions</i>							
Year 1	Year 5	Year 10	Year 11	Year 15	Year 20	Year 25	Year 30
\$5,300	\$29,877	\$69,858	<i>Contributions cease</i>	\$93,486	\$125,105	\$167,419	\$224,045
<i>Neil—\$5,000 annual contribution beginning in year 11 for 25 years; \$125,000 total contributions</i>							
Year 1	Year 5	Year 10	Year 11	Year 15	Year 20	Year 25	Year 30
\$0	\$0	\$0	<i>Contributions begin</i>	\$29,877	\$69,858	\$123,363	\$194,964
<i>The numbers in this chart assume all contributions are made at once at the beginning of each year; 6 percent interest is calculated and applied at the end of each year.</i>							

Recommendations

- ◆ Work toward saving at least 20% of income overall, I suggest 30%.
 - ◆ 12 to 15% toward retirement savings
 - ◆ Remainder toward emergency, planned, and other expenses.
- ◆ Start now!
 - ◆ Save any amount
 - ◆ Then increase from there
- ◆ Recognize psychological side of money
 - ◆ Try to avoid present bias (the desire for things now versus payoff in the future).

Questions?



Next Week: Different types of retirement plans....